Chapter 37 — Data Export, Import & Backup

> **Part IV — Offline-First & Synchronization** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 24 (encryption — what to exclude from backup), Ch 25 (secure files/SAF — export destinations), Ch 3 (Room), Ch 20/22 (secrets/session), Ch 32 (sync — the cloud alternative to device backup). Completes **Part IV**.
>
> **Reference surface in StreamCart:** Android auto-backup of settings/preferences, user-initiated **data export** (GDPR portability), import/restore, and device-to- device transfer — all without leaking secrets.

37.1 Problem Definition and Real-World Context

Users expect their data to **survive** reinstalls and device changes, and regulations (GDPR/CCPA) grant them the right to **export** (data portability) and **delete** their data. Android provides **auto-backup** (to Google Drive) and **device transfer**; apps add **user-initiated export/import**. Each is easy to get dangerously wrong:

- **Backing up secrets.** Android **auto-backup is on by default** and copies app data (SharedPreferences, databases, files) to the cloud — **including tokens, encryption keys, and PII** unless you exclude them. A refresh token in a cloud backup is an account-takeover vector (Ch 19/20); a device-bound Keystore key in a backup is useless on the new device (Ch 20) but the *data it protected* may leak if not excluded.
- **Restoring stale/incompatible data.** A backup restored onto a new app version can carry an old schema (Ch 36) or a key that no longer decrypts (Ch 20/24) → crash or corruption if restore isn't handled.
- **No export for portability.** GDPR requires giving users their data in a portable format; apps without an export feature are non-compliant.
- **Unvalidated import.** Importing a user-supplied file without validation risks corruption or injection (Ch 25).
- **Over-backup.** Backing up large caches wastes the user's backup quota and bandwidth (caches are regenerable — exclude them).

The architecture: configure **auto-backup rules** (what to include/exclude — exclude secrets/caches), handle **restore** (schema/version/key mismatches), provide **user-initiated export** (portable, encrypted where sensitive) and **import** (validated), and know when **cloud sync** (Ch 32) is the better durability mechanism than device backup. This is about **what leaves the device, safely, and what comes back correctly**.

Three practical use cases

1. **Settings/preferences (StreamCart).** Auto-backup DataStore preferences (theme, non-sensitive config) so they survive reinstall — **excluding** the session/token files.
2. **Data export (GDPR).** A user exports their orders/wishlist/profile as a portable file (JSON/ZIP) via SAF (Ch 25), encrypted if sensitive.
3. **Device transfer (OTT).** Moving to a new phone transfers watch-history and settings (via sync, Ch 32) while **re-authenticating** (secrets aren't transferred).

Why this is architectural

Backup/export defines **what app data leaves the device and how it returns** — the boundary between durability/portability and **secret/PII leakage**, the **restore- compatibility** contract (schema/keys/versions), and the **compliance** story (export/delete) — properties that determine data safety and legal compliance.

37.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Auto-backup non-sensitive, non-regenerable data; **exclude** secrets/keys/caches.
F2	Handle restore: schema/version/key mismatches without crash/corruption.
F3	Provide user-initiated **export** in a portable format (GDPR).
F4	Provide **import** with validation of user-supplied data.
F5	Support device transfer; **re-authenticate** (don't transfer secrets).
F6	Support data **deletion** (right to erasure) across device + backend.

Non-functional requirements

Attribute	Policy
Security	No secrets/keys/PII in backups/exports unless encrypted + intended.
Reliability	Restore is safe across versions; import validated.
Compliance	GDPR/CCPA export + delete; consent.
Efficiency	Don't back up regenerable caches (quota/bandwidth).
Portability	Open, documented export format.

Scope

- **In scope:** Android Auto Backup (full-data + key/value), backup rules (include/ exclude), restore handling, user-initiated export/import (format, validation, encryption), device transfer, deletion, sync-vs-backup.
- **Out of scope:** encryption mechanics (Ch 24), file/SAF mechanics (Ch 25), schema migration (Ch 36), cloud sync internals (Ch 32), session/secret storage (Ch 20/22).

Assumptions & constraints

- Android Auto Backup (on by default; `android:dataExtractionRules` on API 31+, `android:fullBackupContent` earlier). SAF for export (Ch 25). Secrets in Keystore (Ch 20). `minSdk 24`. **Auto-backup is opt-out, so exclusion is mandatory.**

37.3 System Design Fundamentals

- **Android Auto Backup (full-data).** By default, Android backs up most app data (files, databases, SharedPreferences, DataStore) to the user's Google Drive (quota-limited, encrypted in transit/at rest by Google) and restores it on reinstall/ new device. **It's on by default** — you control it with **backup rules** (`include`/`exclude` by domain: file/database/sharedpref/root). **Key/Value Backup** (Backup API) is an older, explicit alternative for small data.
- **Exclusion is the critical control. Exclude**: secrets/tokens (even if you think they're encrypted — a device-bound-key ciphertext is useless elsewhere, but don't rely on backup for it; Ch 20), encryption-key material, and **regenerable caches** (quota waste). **Include**: durable non-sensitive user config/state that's cheap to keep and expensive to lose.
- **Restore handling.** Restored data may be from an **older app version** → run schema migrations (Ch 36); a restored **encrypted** file whose **device-bound key** didn't transfer is **undecryptable** (by design, Ch 20/24) → treat as absent and re-provision. Design for "data may arrive stale/incompatible."
- **User-initiated export (portability/GDPR).** Serialize the user's data (orders, profile, wishlist) to a **portable, documented format** (JSON, or a ZIP of JSON + media) and let the user save it via **SAF** (Ch 25). Encrypt if sensitive; the user controls the destination.
- **Import.** Reading a user-supplied export: **validate** (schema/version/size/ integrity), map tolerantly (Ch 16), and merge into the local store — treat as **untrusted input** (Ch 25).
- **Device transfer.** Android's device-transfer (and backup) can move app data, but **secrets don't transfer** (Keystore keys are device-bound) — the user re-authenticates (Ch 19); user data is better moved via **cloud sync** (Ch 32), which is device-independent.
- **Deletion (right to erasure).** Support deleting the user's data on **device** (clear DB/files/outbox, Ch 22) **and** request backend deletion — compliance isn't just local.
- **Backup vs sync. Cloud sync** (Ch 32) is usually the better durability/portability mechanism (device-independent, real-time, server-authoritative); **device backup** is a convenience for local-only/offline config. Prefer sync for important user data; use backup for settings and as a safety net.

Core insight: **auto-backup is on by default — exclude secrets and caches; handle restore as possibly-stale/incompatible; provide validated, portable export/import for GDPR; don't transfer secrets (re-auth); and prefer cloud sync for important data, backup for settings.** The whole game is controlling *what leaves the device* and *what safely returns*.

37.4 High-Level Architecture


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Components & ownership

- **Backup rules** decide what auto-backup includes/excludes (secrets/caches out).
- **Restore handling** migrates schema (Ch 36) and re-provisions secrets (Ch 20).
- **Export/import** serialize/validate portable user data via SAF (Ch 25).
- **Sync** (Ch 32) is the preferred durability path for important data.

Direction: data → backup rules (exclude secrets/caches) → Drive → restore (migrate/re-provision); export → portable file; import → validate → merge.

37.5 Detailed Component Design



Component	Responsibility	Note
backup rules (XML)	include/exclude per domain	exclude secrets + caches
restore handling	schema migrate + secret re-provision	Ch 36/20
`DataExporter`	serialize portable bundle	JSON/ZIP; encrypt sensitive
`DataImporter`	validate + merge	untrusted input (Ch 25)
deletion	device + backend erasure	GDPR (Ch 22)

37.6 End-to-End Data Flow

Auto-backup + restore (secrets excluded)



User export (GDPR) via SAF



37.7 Complete Kotlin / Config Implementation

Package: `com.streamcart.core.backup`.

37.7.1 Backup rules — exclude secrets and caches (mandatory)

```
<!-- res/xml/data_extraction_rules.xml (API 31+); referenced in AndroidManifest -->
<data-extraction-rules>
  <cloud-backup>
    <!-- Include durable, non-sensitive config/data. -->
    <include domain="file" path="datastore/settings.preferences_pb"/>
    <!-- EXCLUDE secrets, keys, and regenerable caches. -->
    <exclude domain="file"     path="datastore/session.pb"/>        <!-- tokens/session (Ch 20)
-->
    <exclude domain="file"     path="datastore/secure.preferences_pb"/>  <!-- wrapped keys (Ch
24) -->
    <exclude domain="database" path="wallet.db"/>                   <!-- encrypted sensitive DB
(Ch 24) -->
    <exclude domain="file"     path="http/"/>                       <!-- HTTP cache
(regenerable) -->
    <exclude domain="file"     path="image_cache/"/>                <!-- image cache (Ch 56) --
>
    <exclude domain="sharedpref" path="legacy_prefs.xml"/>
  </cloud-backup>
  <device-transfer>
    <exclude domain="file" path="datastore/session.pb"/>            <!-- re-auth on new device
-->
    <exclude domain="database" path="wallet.db"/>
  </device-transfer>
</data-extraction-rules>
```

```
<!-- AndroidManifest.xml -->
<application
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules_legacy"   <!-- pre-API-31 equivalent -->
    android:allowBackup="true">  <!-- keep true, but control content precisely via rules -->
```

37.7.2 Restore handling (schema migrate + re-provision secrets)

```
// On first launch after restore: schema is migrated by Room (Ch 36) when the DB opens.
// Secrets were excluded -> absent -> treat as logged out and re-authenticate.
suspend fun onAppStartAfterPossibleRestore(session: SessionManager, keys: KeyManager) {
    // Room opens + migrates automatically (Ch 36). Session derives from token presence (Ch 22).
    if (keys.hasUsableKey().not() || session.current() == null) {
        session.logoutLocalOnly()        // clear any partial state; route to login (Ch 19/22)
    }
}
```

37.7.3 User-initiated export (portable bundle)

```
@Serializable
data class ExportBundle(
    val version: Int = 1,                // for import compatibility (Ch 16)
    val exportedAt: Long,
    val profile: ProfileExport,
    val orders: List<OrderExport>,
    val wishlist: List<WishlistExport>,
)

class DataExporter @Inject constructor(private val repos: ExportRepos, private val json: Json) {
    suspend fun export(): ByteArray {                        // portable, documented format
        val bundle = ExportBundle(
            exportedAt = System.currentTimeMillis(),
            profile = repos.profile().toExport(),
            orders = repos.orders().map { it.toExport() },
            wishlist = repos.wishlist().map { it.toExport() },
        )
        return json.encodeToString(bundle).toByteArray()     // (zip + encrypt if
sensitive/large, Ch 24/25)
    }
}

// UI: write via SAF so the user picks the destination (Ch 25).
val createExport =
registerForActivityResult(ActivityResultContracts.CreateDocument("application/json")) { uri ->
    uri?.let { lifecycleScope.launch { contentResolver.openOutputStream(it)?.use { os ->
os.write(exporter.export()) } } }
}
// createExport.launch("streamcart-export.json")
```

37.7.4 Import with validation (untrusted input, Ch 25)

```
class DataImporter @Inject constructor(private val json: Json, private val repos: ExportRepos) {
    suspend fun import(bytes: ByteArray): ImportResult {
        val bundle = try {
            json.decodeFromString<ExportBundle>(bytes.decodeToString())   // tolerant parse (Ch
16)
        } catch (e: Exception) { return ImportResult.Invalid("unreadable or wrong format") }

        if (bundle.version > CURRENT_IMPORT_VERSION) return ImportResult.Invalid("newer format;
update app")
        // Validate sizes/fields; treat as UNTRUSTED (Ch 25). Then merge (don't blindly
overwrite).
        repos.mergeImported(bundle)                                        // conflict-aware (Ch
33)
        return ImportResult.Success(bundle.orders.size + bundle.wishlist.size)
    }
    private companion object { const val CURRENT_IMPORT_VERSION = 1 }
}
```

37.7.5 Deletion (right to erasure — device + backend)

```
suspend fun deleteAllUserData(session: SessionManager, api: AccountApi) {
    runCatching { api.requestAccountDeletion() }      // backend erasure (GDPR)
    session.logout(allDevices = true)                 // clear local DB/files/outbox/secrets (Ch
22)
}
```

37.7.6 Module structure

```
:core:backup   DataExporter, DataImporter, RestoreHandler, deletion
res/xml        data_extraction_rules.xml, backup_rules_legacy.xml
# Secrets excluded (Ch 20); sync preferred for important data (Ch 32)
```

37.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Backup during a write	inconsistent snapshot	Android backs up at safe points; keep data transactional (Ch 3)
Restore + migration on open	crash/corruption	Room migrates (Ch 36); handle absent secrets
Import merges concurrent edits	overwrite	conflict-aware merge (Ch 33); don't blind-overwrite
Export during active writes	partial export	snapshot/read in a transaction
Restored encrypted data, key absent	undecryptable	treat as absent; re-provision (Ch 20/24)
Deletion vs in-flight sync	partial delete	logout cancels sync; backend erasure is authoritative

- **Restore is the risky path:** restored data may be a stale snapshot from an older version with absent device-bound secrets — migrate (Ch 36) and re-provision/ re-auth rather than assuming it's current and complete.

37.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Secret in backup	security review	add exclusion rule; rotate leaked secret	closed
Restore crashes (old schema)	crash on open	Room migration (Ch 36) handles it	migrated
Restored encrypted data undecryptable	decrypt fails (Ch 24)	treat as absent; re-sync/re-auth	recovered
Import of corrupt/hostile file	validation fails	reject with message; don't merge	safe
Backup quota exceeded	backup skipped	exclude caches; minimize backup size	fits
Device transfer loses secrets	by design	re-authenticate (Ch 19)	expected
User requests deletion	—	device clear + backend erasure	compliant
Over-backup (large caches)	quota/bandwidth	exclude regenerable caches	efficient


37.9 Failure Scenarios and Recovery

> **Assume restored data is stale and secrets are gone.** Migrate the schema, re-authenticate, and reconcile via sync (Ch 32) — never assume a restore gives you a current, complete, decryptable state.

37.10 Scaling the Implementation

- **Level 1.** Configure backup rules: include settings, **exclude secrets + caches**; `allowBackup=true` controlled by rules.
- **Level 2.** Restore handling (migrate, re-auth); basic export (JSON via SAF).

- **Level 3 — production.** Validated import (conflict-aware merge); encrypted export for sensitive data; deletion (device + backend); device-transfer rules.
- **Level 4 — large.** Prefer **cloud sync** (Ch 32) for important data (backup as safety net); ZIP export with media; versioned export format (Ch 16); background export for large datasets.
- **Level 5 — enterprise.** Full GDPR/CCPA tooling (export/delete/consent), data-residency, audit trails, encrypted end-to-end backup, cross-platform portable formats.

Avoid overengineering: most apps need correct backup rules + a simple export + deletion. Cloud sync (Ch 32) handles durability/portability for important data better than device backup — don't over-invest in device backup for data you sync.

37.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Durability	Cloud sync (Ch 32)	Android device backup	Sync for important/portable data (device-independent); backup for local settings/safety net
Backup type	Auto Backup (full-data)	Key/Value (Backup API)	Auto for most; K/V for small explicit data/legacy
Secrets	Exclude (re-auth on restore)	back up encrypted	**Exclude** — device-bound keys don't transfer; re-auth
Export format	Open JSON/ZIP (portable)	proprietary	Open/documented for GDPR portability
Import	Validate + conflict-merge	blind overwrite	Validate + merge (untrusted input, conflicts)
Caches	Exclude (regenerable)	back up	Exclude — quota/bandwidth waste

Sync vs device backup: cloud sync (Ch 32) keeps data on the server, available on any device in real time, authoritative and access-controlled — the better choice for important user data. Android device backup is a convenient, zero-effort safety net for **local-only** settings and offline state, but it's a point-in-time snapshot, can carry secrets if misconfigured, and is tied to the backup transport. **Sync the important stuff; back up settings.**

Exclude vs encrypt-and-backup secrets: excluding secrets is the safe default — device-bound Keystore keys (Ch 20) don't transfer anyway, so a backed-up ciphertext is useless on a new device, and backing up the secret risks exposure. Re-authenticate on restore instead. Only back up (encrypted) secrets if you have a device-independent key-escrow scheme, which is advanced and risky.

37.12 Security Considerations

- **Exclude all secrets/keys/PII from backup** (the #1 rule) — tokens (Ch 19), Keystore-wrapped keys (Ch 20), sensitive DBs/files (Ch 24/25). Auto-backup is **on by default**, so forgetting to exclude is a leak.
- **Exports of sensitive data must be encrypted** (Ch 24) and user-controlled (SAF, Ch 25) — a plaintext export of personal data on shared storage leaks.

- **Import is untrusted input** (Ch 25) — validate format/size/integrity; never execute or blindly trust; tolerant parse (Ch 16).
- **Device-bound keys don't transfer** — re-authenticate on restore; don't attempt to back up Keystore material.
- **Right to erasure** (GDPR) — delete on device **and** request backend deletion; don't leave PII in backups you can't reach.
- **Backup transport** is Google-encrypted, but defense-in-depth: don't rely on it for secrets.
- Client vs backend: backups/exports handle **local** data; the backend owns the authoritative copy, access control, and server-side deletion.

37.13 Performance and Resource Optimization

- **Exclude regenerable caches** (HTTP/image/DB caches) — biggest win for backup quota/bandwidth.
- **Minimize backup size:** back up only durable, small, non-sensitive state; sync the rest (Ch 32).
- **Background/stream large exports:** a big ZIP export should stream (Ch 25), not buffer in memory; run off-main.
- **Restore cost:** migrations run at open (Ch 36/88) — keep fast.
- **Profiling:** backup size, backup frequency/skips (quota), export/import duration, restore-time migration cost.

37.14 Testing Strategy

Backup rules and restore are hard to unit-test; test export/import logic in isolation and backup rules via static checks + manual/instrumentation.

```
class ExportImportTest {
    @Test fun `export is a portable readable bundle`() = runTest {
        val bytes = exporter.export()
        val bundle = json.decodeFromString<ExportBundle>(bytes.decodeToString())
        assertEquals(1, bundle.version); assertTrue(bundle.orders.isNotEmpty())
    }
    @Test fun `import round-trips exported data`() = runTest {
        val bytes = exporter.export(); val result = importer.import(bytes)
        assertTrue(result is ImportResult.Success)
    }
    @Test fun `import rejects corrupt file`() = runTest {
        assertTrue(importer.import("not json".toByteArray()) is ImportResult.Invalid)
    }
    @Test fun `import rejects newer format`() = runTest {
        val future = json.encodeToString(ExportBundle(version = 99, exportedAt = 0, /*...*/))
        assertTrue(importer.import(future.toByteArray()) is ImportResult.Invalid)
    }
    // Static/CI check: assert backup rules EXCLUDE secret paths.
    @Test fun `backup rules exclude session and secrets`() {
        val xml = readRes("data_extraction_rules.xml")
        assertTrue(xml.contains("exclude") && xml.contains("session.pb"))
        assertTrue(xml.contains("secure.preferences_pb"))
    }
}
```

Manual/instrumentation: backup+restore via `adb shell bmgr` (force backup/restore); verify secrets absent after restore and re-auth required; restore onto a new app version (schema migration, Ch 36).

37.15 Production Observability

- **Metrics:** backup size, backup success/skip (quota) rate, restore rate, restore- time migration success (Ch 36), export/import counts + failure rate, deletion completion rate.
- **Logs:** `backup{size}`, `restore{migrated, reauth}`, `export{items}`, `import{result}` (no data contents, Ch 28).
- **Debugging:** secret found in a backup → missing exclusion (fix + rotate); crash-on-restore → schema not migrated (Ch 36); import failures → format/validation; backup skipped → quota (exclude caches); users re-logged-in after restore → expected (secrets excluded).
- **Alerting:** a secret-in-backup finding → security incident; restore-crash spike → migration issue.
- **Privacy:** exports/backups are user data — never log contents; support deletion.

37.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Is Android auto-backup on by default?* Yes — it copies app data to Drive unless you exclude it.
2. *What must you exclude?* Secrets/tokens, encryption keys, sensitive DBs/files, and regenerable caches.
3. *Why exclude secrets?* Leak risk (account takeover); device-bound keys don't transfer anyway (Ch 20).
4. *Auto Backup vs Key/Value Backup?* Full-data (default, most data) vs small explicit key/value (legacy).
5. *How configure?* `android:dataExtractionRules` (API 31+) / `fullBackupContent` with include/exclude by domain.

Restore 6. *Restore risks?* Stale/old-version data; absent device-bound secrets. 7. *Handle old schema?* Room migrations run on open (Ch 36). 8. *Absent secrets after restore?* Re-authenticate (Ch 19); treat as logged out. 9. *Restored encrypted data, key gone?* Undecryptable by design; re-provision/re-sync.

Export/import (GDPR) 10. *Why export?* Data portability (GDPR/CCPA). 11. *Format?* Open, documented, versioned (JSON/ZIP); encrypt if sensitive. 12. *Where?* User-chosen via SAF (Ch 25). 13. *Import safety?* Validate (untrusted input, Ch 25); tolerant parse (Ch 16); conflict-aware merge (Ch 33). 14. *Right to erasure?* Delete device data + request backend deletion (Ch 22).

Device transfer 15. *Do secrets transfer?* No — Keystore keys are device-bound; re-auth. 16. *Better durability for important data?* Cloud sync (Ch 32), device-independent.

Backup vs sync 17. *When backup?* Local-only settings/offline state; safety net. 18. *When sync?* Important user data (authoritative, real-time, portable).

Security 19. *Sensitive export?* Encrypt (Ch 24) + user-controlled destination. 20. *Caches in backup?* Exclude (quota/bandwidth). 21. *Transport security?* Google-encrypted, but don't rely on it for secrets.

Performance 22. *Reduce backup size?* Exclude caches; back up only durable small state. 23. *Large export?* Stream (Ch 25); background.

Testing 24. *How test export/import?* Round-trip; reject corrupt/newer-format. 25. *How verify exclusions?* Static check rules; `adb bmgr` backup/restore; assert secrets absent.

Production debugging 26. *Secret in backup?* Missing exclusion. 27. *Crash on restore?* Schema not migrated. 28. *Backup skipped?* Quota (caches not excluded).

Design-change challenges

1. **Token leaked via auto-backup.** Add exclusion; rotate; re-auth on restore.
2. **GDPR export request.** Portable JSON/ZIP via SAF; encrypt sensitive.
3. **Crash after restore on new app version.** Room migration (Ch 36).
4. **Move data to a new phone.** Cloud sync (Ch 32) + re-auth; secrets not transferred.
5. **Right to be forgotten.** Device clear + backend deletion.

Mock walkthrough (abridged)

Clarify (what data, secrets, compliance, device change?) → auto-backup rules (exclude secrets + caches) → restore handling (migrate Ch 36, re-auth) → export (portable/encrypted via SAF) + validated import → deletion (device+backend) → prefer sync (Ch 32) for important data → security (no secret leak) → testing (`bmgr`, round-trip).

37.17 Practical Exercises

1. **Beginner.** Configure backup rules to include settings and **exclude** the session/secrets and caches; verify via `adb bmgr`. *Eval:* secrets absent after restore.
2. **Intermediate.** Handle restore: migrate schema (Ch 36) and re-authenticate when secrets are absent. *Eval:* no crash; user re-logs-in.
3. **Advanced.** Build a portable, versioned export (SAF) + validated import with conflict-aware merge. *Eval:* round-trips; rejects corrupt/newer.
4. **Production debugging.** Fix a token leaking into auto-backup and a crash-on- restore (old schema). *Eval:* excluded + rotated; migrated.
5. **System design challenge.** Implement GDPR export + deletion (device + backend) with encrypted sensitive export. *Eval:* compliant; no PII left; encrypted.

(Solutions withheld unless requested.)

37.18 Chapter Summary & Reference Sheet

Key decisions: Android **auto-backup is on by default — exclude secrets/keys/PII and regenerable caches** via backup rules (include only durable non-sensitive state); **handle restore as possibly-stale/incompatible** (migrate schema Ch 36, re-authenticate since device-bound secrets don't transfer, Ch 20); provide **user-initiated, portable, versioned export** (JSON/ZIP via SAF, encrypted if sensitive, Ch 24/25) and **validated import** (untrusted input, conflict-aware merge Ch 33); support **deletion** (device + backend, GDPR); **prefer cloud sync (Ch 32)** for important data, device backup for settings/safety net. The game is controlling **what leaves the device safely** and **what returns correctly**.

Concepts: Android Auto Backup / Key-Value Backup, backup rules (include/exclude by domain), restore handling (migrate/re-provision), data portability/export, validated import, device transfer, right to erasure, sync-vs-backup.

Common mistakes: backing up secrets/tokens/keys (default-on leak); backing up regenerable caches (quota); assuming restore gives current/complete/decryptable data; no schema migration on restore (crash); proprietary/unversioned export; blind import (corruption/injection); no deletion support (non-compliant); over-relying on device backup instead of sync.

Checklist: ☐ backup rules exclude secrets+caches ☐ `allowBackup` controlled by rules ☐ restore: migrate (Ch 36) + re-auth ☐ portable versioned export (SAF, encrypt sensitive) ☐ validated conflict-aware import ☐ deletion (device+backend) ☐ prefer sync (Ch 32) for important data ☐ test with `adb bmgr` + round-trip ☐ no data in logs.

Key APIs/config: `android:dataExtractionRules` / `fullBackupContent`, `<cloud-backup>` / `<device-transfer>` include/exclude, `android:allowBackup`, `ActivityResultContracts.CreateDocument` / `OpenDocument` (SAF, Ch 25), `adb shell bmgr`, `MigrationTestHelper` (restore schema, Ch 36), encryption (Ch 24).

References:

- Back up user data (Auto Backup): https://developer.android.com/guide/topics/data/autobackup
- Include/exclude backup rules: https://developer.android.com/guide/topics/data/autobackup#IncludingFiles
- Data portability / privacy: https://developer.android.com/guide/topics/data/collect-share
- (composes) Ch 24 Encryption, Ch 25 Files/SAF, Ch 32 Sync, Ch 36 Migrations, Ch 22 Session/Deletion

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; backup rules + restore handling + export/import + deletion; concurrency (restore risk, import merge); failure (secret leak, restore crash, corrupt import); 5 levels; sync-vs-backup and exclude-vs-encrypt trade-offs; security (exclude secrets, encrypt export); perf (exclude caches); export/import + rules tests + `bmgr`; observability (secret-in-backup alert); 28 Q&A + 5 challenges. ✅
- **Expand later:** cloud sync (Ch 32) as primary durability, GDPR tooling, encrypted end-to-end backup, key escrow.
- **Interview coverage:** exceeds ≥25 bar. **Completes Part IV (Ch 29–37).**